

PROJECT TITAN

Engineering Calculations and FEA Sanity Check

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EDUCATIONAL USE NOTICE

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Document Control

Revision	Date	Description	Author
A	July 2026	Initial analytical sanity-check release based on confirmed geometry and load inputs.	Michael Lozada-Longog

1. Purpose and Scope

This calculation book performs first-order checks that can be supported by the available geometry. It does not attempt to replace the full assembly FEA. Because the bearing loads act through bearing centerlines and the complete support geometry is complex, a unique shaft-bending solution cannot be derived without a complete free-body diagram of the external supports and load direction.

2. Confirmed Inputs

Parameter	Value
Shaft diameter d	10 mm
Shaft length	40 mm
Bearing center spacing L	31 mm
Bearing inner diameter	10 mm
Bearing outer diameter	22 mm
Bearing width b	7 mm
Plate/frame thickness t	6 mm
Load at each bearing P	2,000 N
Total applied load	4,000 N
FEA max stress	54.947 MPa
FEA min safety factor	2.377
FEA max displacement	0.004 in = 0.1016 mm

3. Bearing Center Spacing Derivation

The Measure tool reported 21 mm between the two selected inner circular edges. Each selected edge has radius 5 mm. Therefore center-to-center spacing is:

$$L = 21 + 5 + 5 = 31 \text{ mm}$$

4. Load Accounting

Two independent bearing loads were configured: $P_1 = 2,000 \text{ N}$ and $P_2 = 2,000 \text{ N}$. The total applied load magnitude is 4,000 N. For local checks, each bearing is evaluated using 2,000 N; loads are not combined into a single 4,000 N force at one bearing.

5. Shaft Transverse Shear Check

Shaft cross-sectional area $A = \pi d^2 / 4 = 78.54 \text{ mm}^2$.

Average shear stress per 2,000 N load: $\tau_{avg} = V/A = 25.46 \text{ MPa}$.

Maximum shear stress for a solid circular section under transverse shear: $\tau_{max} = 4V/(3A) = 33.95 \text{ MPa}$.

This is a local first-order shear check. The actual shaft stress state may include bending, contact, and constraint effects not represented here.

6. Bearing Projected Pressure Checks

Inner-race projected pressure using shaft diameter x bearing width: $p = 2000/(10 \times 7) = 28.57 \text{ MPa}$.

Outer-race projected pressure using bearing OD x width: $p = 2000/(22 \times 7) = 12.99 \text{ MPa}$.

Plate projected bearing pressure using bearing OD x plate thickness: $p = 2000/(22 \times 6) = 15.15 \text{ MPa}$.

These projected-area values are screening calculations, not Hertzian-contact solutions. Detailed contact pressure requires bearing geometry, race curvature, clearances, and material properties.

7. FEA Material-Model Consistency Audit

Fusion reports maximum von Mises stress = 54.947 MPa and minimum safety factor = 2.377. These values imply an effective yield strength of approximately $54.947 \times 2.377 = 130.6 \text{ MPa}$.

This implied value does not match the commonly cited nominal yield strength of 6061-T6 aluminum used in the design intent. Therefore, the exact material assigned in Fusion must be checked before claiming that the FEA safety factor represents 6061-T6. The result itself remains valid for the material model actually used, but the material name/yield basis must be confirmed.

8. Comparison and Interpretation

Check	Analytical result	FEA/Reference	Interpretation
Shaft maximum transverse shear per bearing	33.95 MPa	Assembly max von Mises 54.947 MPa	Different stress measures and locations; order of magnitude is plausible
Bearing inner projected pressure	28.57 MPa	No direct equivalent	Useful local screening only
Plate projected pressure	15.15 MPa	Stress concentration around interfaces	Supports local interface review
Maximum displacement	Not uniquely derivable from current simplified geometry	0.1016 mm	Requires complete support/load free-body model
Safety factor	Material-dependent	2.377	Material assignment audit required

9. Conclusions

- The confirmed geometry and load accounting are internally documented.
- Simple shear and projected-pressure checks do not indicate an obviously impossible FEA magnitude.

- A defensible closed-form bending/deflection calculation requires a fully defined external load path and support distances, which are not presently available.
- The Fusion material assignment is the highest-priority audit item because the reported stress and safety factor imply an effective yield of about 130.6 MPa.

10. Recommended Next Verification

Capture screenshots of each component material property in Fusion, document the exact constraint surfaces and load directions, and create a simplified free-body diagram with distances from bearing centerlines to fixed support rows. Those data will permit a true beam/plate correlation calculation in Revision B.